



UTM
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SECP3213 – BUSINESS INTELLIGENCE

SECTION 01

PHASE 2: DATA WORKFLOW DOCUMENTATION

Strategic Real Estate Investment Intelligence: A Multi-Dimensional Analytics Solution for Property Valuation and Market Opportunity Mapping

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1.0 Workflow Architecture and Tools Used

1.1 Workflow Architecture

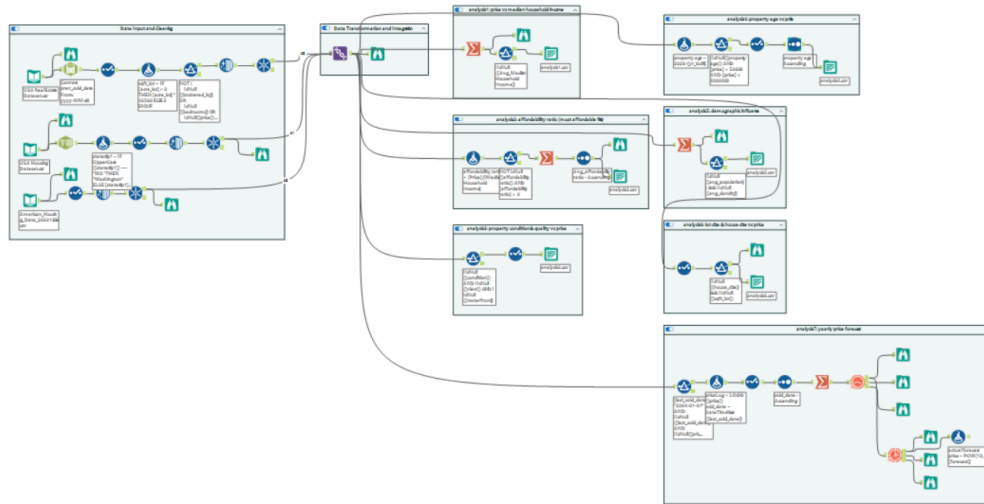


Figure 1: Overall Workflow of the Architecture

Moving from left to right across the canvas, the workflow follows a logical sequence. Container 1 handles the ingestion and cleaning of three raw datasets obtained from Kaggle. From there, Container 2 integrates the cleaned datasets using a Union tool. Finally, Containers 3 through 9 represent seven parallel analytical branches, each performing a distinct business intelligence calculation.

1.2 Data Input and Cleaning

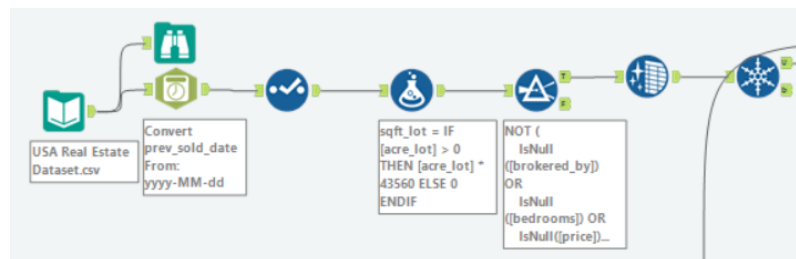


Figure 2: Cleaning Process for USA Real Estate Dataset

A DateTime tool was applied to convert this column to a standard date format. Next, a Select tool was used to modify data types for several columns' data types. A Formula tool was then introduced to create a new column named sqft_lot as the source dataset provides lot size in acres, but the other datasets use square feet. To make comparisons possible, conversion was required. The formula applied was: IF [acre_lot] > 0 THEN [acre_lot] * 43560 ELSE 0 ENDIF. Where acre_lot values were missing or zero, the formula simply returned zero. Following this, a Filter tool was configured to remove any record containing null values in critical columns. After filtering, a Data Cleansing tool was applied to remove leading and trailing whitespace from all string columns. Finally, a Unique tool was used to remove duplicate rows based on all columns.

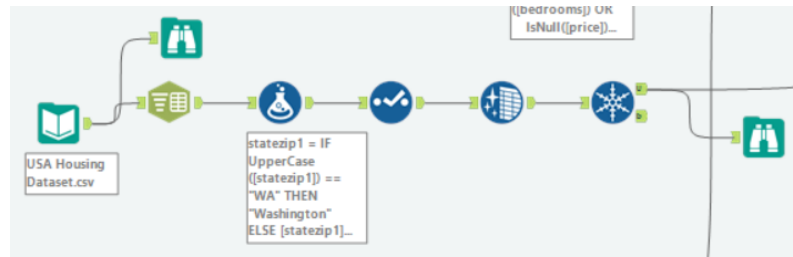


Figure 3: Cleaning Process for USA Housing Dataset

Moving on to the second dataset, a specific challenge became apparent. The statezip column combined state abbreviations and zip codes in a single field, with values like "WA 98101". To work with this data effectively, state and zip code need to be separate columns. A Text to Columns tool was configured to split this column based on the space delimiter. A Formula tool was then applied to convert state abbreviations to full state names. Specifically, the tool checked whether the value in statezip1 equaled "WA" and, if true, replaced it with "Washington". This conversion was necessary because the other two datasets use full state names rather than abbreviations. Following this, a Select tool was used to rename statezip1 to state and statezip2 to zip_code. A Data Cleansing tool then removed leading and trailing whitespace from all columns except state and zip_code, and a Unique tool eliminated any duplicate records.

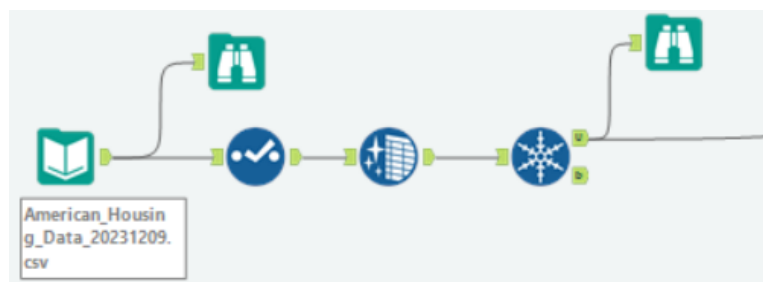


Figure 4: Cleaning Process for American Housing Dataset

This dataset required the least cleaning. A Select tool was used to rename and changed data types of several columns. A Data Cleansing tool removed whitespace from all columns except bedrooms and bathrooms, and this was followed by a Unique tool for duplicate removal.

1.3 Data Transformation and Integration

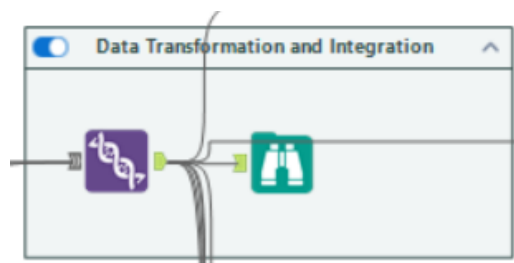


Figure 5: Integration of 3 datasets

With all three datasets cleaned individually, the next task was to combine them into a single master dataset. Container 2 houses the Union tool, which stacks all three cleaned datasets vertically. The tool was configured to automatically match columns by name. This meant that when a column was present

in only one dataset, the Union tool retained the column and simply left null values for records coming from other datasets. Our team chose this approach because it preserved the richness of the data while still establishing a unified structure.

1.4 Analytical Techniques

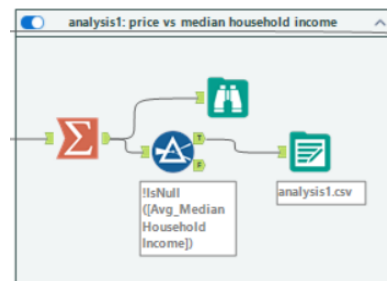


Figure 6: Price vs Median Household Analysis Workflow

The first analytical container compares average house prices against median household income at the state level. The integrated dataset was fed into a Summarize tool configured to group by state and calculate both the average price and the average median household income for each state. After this aggregation, a Filter tool removed any state where the average median household income was null, as such records would have made the comparison invalid.

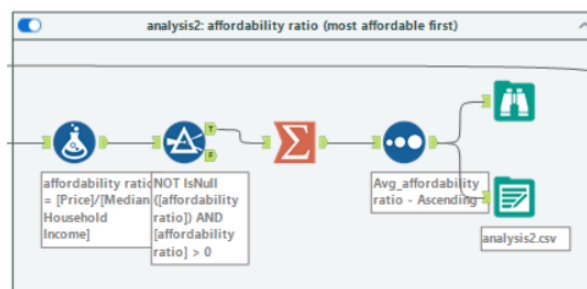


Figure 7: Affordability Ratio Workflow

This analysis was designed to identify geographic areas where housing is most affordable relative to local incomes. The team began by adding a Formula tool to create a new column named affordability ratio. The calculation was simple: Price divided by Median Household Income. The resulting ratio represents the number of years of household income required to purchase a property in a given area. Lower ratios indicate better investment opportunities. A Filter tool was then used to exclude records where the affordability ratio was null, zero, or negative. After this cleaning step, a Summarize tool grouped the data by state and city, calculating the average affordability ratio for each city. A Sort tool arranged the results in ascending order, which placed the most affordable cities at the top of the list.

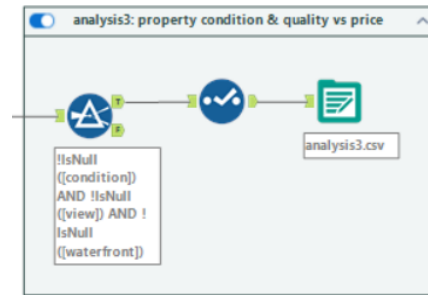


Figure 8: Property Condition & Quality vs Price Analysis Workflow

This analysis examines how property condition, view quality, and waterfront access correlate with sale price. A Filter tool retained only those records containing non-null values for condition, view, and waterfront. It is worth noting that these columns are present exclusively in the USA Housing Dataset, so this analysis necessarily operates on a subset of the integrated data rather than the full dataset. A Select tool then kept only the columns price, city, state, waterfront, view, and condition.

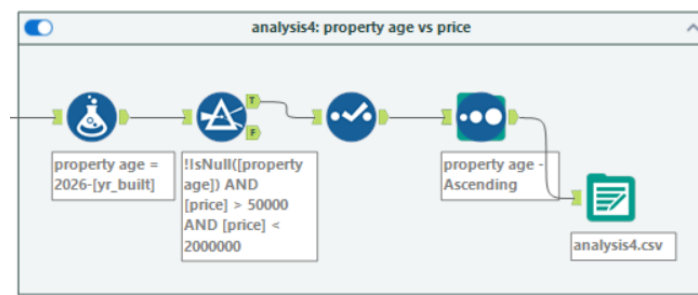


Figure 9: Property Age vs Price Analysis Workflow

The purpose of this analysis was to assess the relationship between property age and sale price. A Formula tool created a column named property age, calculated as 2026 minus yr_built. A Filter tool then removed records with null property ages and applied price boundaries. Specifically, only properties priced between \$50,000 and \$2,000,000 were retained. Properties below \$50,000 were presumed to represent either data errors or land-only sales, while properties above \$2,000,000 were excluded as luxury outliers that would have skewed the analysis. A Sort tool arranged the remaining data in ascending order by property age, placing the oldest properties first.

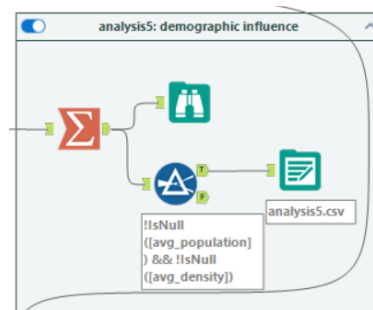


Figure 10: Demographic Influence Analysis Workflow

This analysis investigates how population size and population density affect housing prices. A Summarize tool grouped the integrated dataset by zip_code and then calculated the average price,

average zip code population, and average zip code density for each zip code. A Filter tool removed any records where either average population or average density was null.

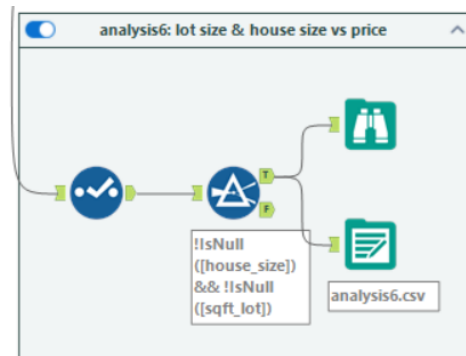


Figure 11: Lot and House Size vs Price Analysis Workflow

This was the most straightforward of the seven analyses. A Select tool retained only the columns price, house_size, and sqft_lot. A Filter tool then excluded records where either house_size or sqft_lot was null.

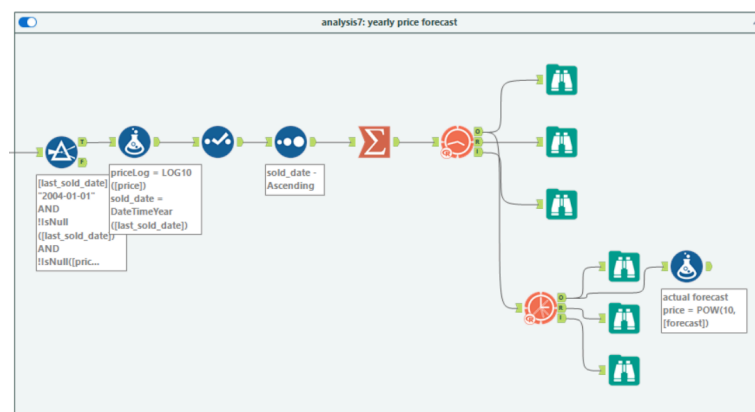


Figure 12: Yearly Price Forecast Workflow

The objective was to forecast housing prices for the next 15 years using time series analysis. To begin, a Filter tool restricted the data to properties sold before January 1, 2004, with prices between \$100,000 and \$1,500,000. The pre-2004 filter ensured at least 20 years of historical data for model training. Next, a Formula tool created a new column named priceLog, calculated as LOG10 of price. A Select tool then retained only priceLog and sold_date. A Sort tool arranged the data in ascending chronological order by sold_date. Following this, a Summarize tool grouped the data by sold_date and calculated the average priceLog for each year. This produced a clean time series with one value per year. Upon ARIMA execution, the model produced an RMSE value of 0.1479. The TS Forecast tool then generated forecasts for 15 periods (years) into the future. Finally, a Formula tool converted the forecast values back to actual dollar amounts. This produced forecasted prices in US dollars, which are far more interpretable for business stakeholders than log-scale values.

2.0 Challenges

1. Combining Inconsistent Datasets

It was difficult to integrate three datasets from different Kaggle sources because of the inconsistent column naming conventions, data type differences across datasets and differences in the unit of measurement in the datasets. In the USA Real Estate Dataset, lot size is measured in acres, while in the USA Housing Dataset, it is measured in square feet, so it has to be converted using a Formula tool before being merged. Moreover, the state and city geographic fields were different in each dataset included abbreviations of the states and other used the full name of the state, so they had to be preprocessed as strings to enable a reliable integrated data set from the Union tool. There were multiple data cleaning and validation steps across all three sources that ensured there was consistency and that the final merged dataset was analytically usable.

2. Missing Values and Price Outliers After Integration

There were many null values in the various columns of the integrated dataset, including price, city, state, and size of the property, and extreme price outliers between \$0 and greater than \$25 million. Removing all rows with some missing data could cause too much data to be lost and could make the analysis results less meaningful and keeping outliers could skew the analysis and visualisations. A balanced approach was used in applying a targeted Filter tools per analysis branch, that is, items with no values (null fields) in the critical fields for that given analysis were removed, and price ranges were capped within a realistic residential price range. This allowed for a sufficient number of data points to be retained for analysis, but also maintained statistical integrity.

3. Optimizing ARIMA Forecasting Model

To set up the ARIMA forecasting model, there are several data preparation challenges specific to time series requirements to overcome. The date field (last_sold_date) used to create the time axis were tens of thousands of individual rows initially, rather than a meaningful series. In order to do this, the year component had to be separated from the data with a Formula tool and then the average price per year had to be calculated prior to input into the ARIMA tool. Further, due to the lack of trend signal, the initial model used was a degenerate ARIMA(0,0,0) model, but this was addressed by manually creating the ARIMA(1,1,1) model and transforming it with a log transformation to stabilise the variance in prices, resulting in MAPE = 2.33% and RMSE = 0.14.